BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan East Gas Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

September 2019

Prepared Rebuttal Testimony of:

Staff Efficiency and
Sustainability Panel

Robert Cully
Utility Engineering Specialist 3

Joseph Hitt
Utility Analyst 2

Laurie Kokkinides
Utility Analyst 1

Kathryn Mammen
Utility Analyst 3
Office of Markets and Innovation

Davide Maioriello
Utility Engineering Specialist 3

Jennifer Park
Assistant Engineer
Office of Electric, Gas and
Water
State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Efficiency and Sustainability Panel (Panel), please state your names.

A. Our names are Robert Cully, Joseph Hitt, Laurie Kokkinides, Kathryn Mammen, Davide Maioriello, and Jennifer Park.

Q. Has the Panel previously sponsored testimony in these proceedings?

A. Yes. We offered pre-filed direct testimony, submitted on August 30, 2019. Our pre-filed direct testimony addressed the Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan East Gas Corporation d/b/a National Grid's (KEDLI), collectively the Companies, proposals related to Energy Efficiency programs; the Utility Energy Services Contract Program; the Fuel Switching Calculator; Demand Response; the Expanded Geothermal Demonstration Projects; the Clean Conversion Program; the Low-and Moderate-Income Conversion Program; renewable natural gas interconnections; the Power-to-Gas Demonstration Project; and the Hydrogen Blending Study.

Q. What is the purpose of the Panel's rebuttal

testimony?

A. The purpose of our rebuttal testimony is to address: (1) the direct testimony of the Environmental Defense Fund (EDF) witness Ferit Ucar with regard to the requirements of the recently passed Climate Leadership and Community Protection Act (CLCPA); (2) the direct testimonies of Estates NY Real Estate Services LLC (Estates) witness Jared Rodriguez, and intervenor Bob Wyman with regard to their proposals to transition the Companies from Natural Gas businesses to “thermal utilities” which would sell heat and cooling services supplied by utility-owned geothermal loop fields; and (3) the City of New York (City) Policy Panel’s (City Policy Panel) proposal to undertake a district heating study and pilot.

Q. Is the Panel sponsoring Exhibits as part of this rebuttal testimony?

A. No.

Climate Leadership and Community Protection Act

Q. Are you familiar with the CLCPA?

A. Yes. It is a New York State law enacted this

year.

Q. How is the CLCPA relevant to these proceedings?

A. Witness Ucar refers to the CLCPA as the basis for certain of his proposals. While the CLCPA's provisions speak for themselves, we address witness Ucar's proposals and the stated rationales for those proposals.

Q. Provide an overview of witness Ucar's proposals with respect to the CLCPA.

A. On pages five and six of his direct testimony, witness Ucar outlines various provisions of the CLCPA with respect to the Act's modifications to New York State's Environmental Conservation Law. Specifically, witness Ucar states that the CLCPA requires New York State to implement measures to reduce statewide greenhouse gas (GHG) emissions levels by 85 percent of the 1990 levels by the year 2050, with up to an additional 15 percent reduction sourced from carbon offsets. On pages six and seven of his direct testimony, witness Ucar also summarizes the Act's requirements for the Public Service Commission, which he describes as including: (1) ensuring that 70 percent of statewide electric generation be

"generated by renewable energy systems" by the year 2030, with 100 percent of statewide electrical demand met by carbon-free generation resources by the year 2040; (2) establishing programs to procure nine gigawatts (GW) of electricity generation from offshore wind resources by the year 2035, six GW of solar photovoltaic generation by the year 2035, and three GW of energy storage capacity by the year 2030; (3) requiring that a minimum amount of the programs implemented by the Commission benefit low and moderate income customers and disadvantaged communities. In addition, on page seven of his direct testimony, witness Ucar summarizes the CLCPA's requirement that all New York State agencies, including the Public Service Commission, consider whether administrative approvals and decisions are inconsistent with attaining statewide emissions limits established in a new addition to the New York State Environmental Conservation Law.

Q. Does witness Ucar provide an opinion as to what he believes the Commission should to do to consider whether its decisions are consistent

with the CLCPA?

A. Yes. On page seven of his direct testimony, witness Ucar contends that the Commission would need a comprehensive current baseline of GHG emissions in each regulated utility service territory. Further, witness Ucar opines that the Commission would need utilities to provide sufficient information to assess actual progress being made in each utility service territory towards GHG emissions reductions.

Q. Does witness Ucar propose specific actions in this rate proceeding related to the CLCPA?

A. Yes. On pages nine and 10 of his direct testimony, witness Ucar proposes that the Companies develop a comprehensive GHG assessment methodology, and that such methodology should include the development of a utility-specific economy-wide GHG emissions inventory and analysis. Witness Ucar states, on page 24 of his direct testimony, that the Companies should develop their own emissions data repositories in a similar fashion to the methodology required of the New York State Department of Environmental Conservation.

Q. Are there important facets of the CLCPA which witness Ucar did not highlight in his testimony?

A. Yes, there are three main areas that witness Ucar did not highlight which deserve further discussion. First, the CLCPA itself defines a process for developing recommendations for how to meet New York State's climate obligations. Second, while the CLCPA did require that the Commission consider whether its decisions would be consistent with meeting the GHG emissions reductions requirements, it did not specify how the Commission must make such discretionary determinations. Third, the CLCPA requires that the New York State Department of Environmental Conservation (NYSDEC), not individual utilities, develop a statewide emissions inventory.

Q. What process does the CLCPA define for developing recommended actions to meet New York State's climate requirements?

A. The CLCPA implements a new section of New York State Environmental Conservation Law, §75, which establishes the New York State Climate Action Council (Climate Action Council), a body of 22 members ranging from the commissioners of

various State agencies, the presidents of various State authorities, the chairperson of the Commission, as well as a number of other appointees designated by the Governor and members of the State Senate and Assembly. The CLCPA requires the Climate Action Council to convene advisory panels and working groups to help it deliberate, and, within two years of the effective date of the Act, prepare and approve a scoping plan outlining recommendations for attaining emissions reductions of 40 percent of 1990 levels by the year 2030 and 85 percent reductions of such levels by 2040, achieving emissions reductions beyond the 85 percent level required by 2040, as well as achieving net-zero emissions from all sectors of the New York State economy. The CLCPA requires that the Climate Action Council's recommendations be incorporated into the next State Energy Plan following the development of such recommendations.

Q. Does the Panel have any specific recommendations related to witness Ucar's proposals?

A. Yes. We have two recommendations.

Q. Please describe your first recommendation.

A. We recommend that the Commission continue to adopt the programs as we recommended in our direct testimony, which are consistent with Commission policy, while the Climate Action Council goes about its work as required by the CLCPA to develop further specific recommendations. That is, the programs which result from this proceeding should be consistent with both Commission policy and the requirements of the CLCPA, but should not pre-judge the recommendations of the Climate Action Council nor make an end-run around the process required by the CLCPA.

Q. Please describe your second recommendation.

A. We recommend rejecting witness Ucar's proposal for comparing carbon emissions reductions to a baseline emissions inventory in a utility's service territory in these proceedings. Witness Ucar's proposal is but one of any number of possible methods the Commission might employ to determine whether its decisions are consistent with the requirements of the CLCPA. Alternatively, the Commission could require the Companies to work closely with the NYSDEC to

ensure that the statewide GHG emissions report and models developed by that agency are sufficiently granular to be useful for utility-specific purposes without requiring significant duplication of efforts. A third option would be to establish a consistent methodology for use across utility service territories. The Commission should decide for itself how to determine whether its actions are inconsistent with the carbon emissions reduction goals set forth in the CLCPA. Thus, such decision is beyond the scope of these rate proceedings, which affect only two utilities and address their rates only for the Rate Year.

Q. Could there be a significant overlap or duplication of effort if each utility in New York State were required to develop and maintain their own service territory-specific GHG inventories?

A. Yes. A KEDNY customer taking Electric service from Con Edison, for example, could be paying for a KEDNY specific inventory, a Con Edison-specific inventory, a New York City-specific inventory, and the statewide inventory.

Gas Utility Transition to Thermal Utility

Q. Provide an overview of Estates witness Rodriguez's direct testimony regarding transitioning the Companies from gas utilities to thermal utilities.

A. On pages nine and 10 of his direct testimony, witness Rodriguez states that the Companies may be able to develop alternative revenue sources through an affiliated thermal company which would sell thermal energy sourced from geothermal and water-sourced loop fields instead of gas. On page 11 of his direct testimony, witness Rodriguez encourages the Commission to investigate allowing regulated monopolies to establish thermal company affiliates. Witness Rodriguez further proposes, on page 15 of his direct testimony, that the Commission provide the Companies with access to additional customers by allowing the Companies to develop new business models such as shared loop geothermal projects and development of a thermal company affiliate. On pages 9 and 11 of his direct testimony, witness Rodriguez states, without elaboration, that there may be legal

hurdles which currently restrict gas utilities from engaging in a line of business selling thermal energy.

Q. What is a thermal company?

A. On page 10 of his direct testimony, witness Rodriguez defines a thermal company as “a company that builds, maintains and manages shared infrastructure … that provides a heating and cooling medium for water source heat pumps or similar technologies.” Witness Rodriguez states that such a thermal company’s distribution facilities would likely be horizontal water pipe networks, heat exchangers, pumps and pumping stations, and physical connections to customers.

Q. Provide an overview of intervenor Wyman’s direct testimony regarding transitioning the Companies from gas utilities to thermal utilities.

A. On page 13 of his direct testimony, intervenor Wyman states that the Companies should invest in non-gas assets, and should “pump heat and not gas.” Intervenor Wyman states that shared district geothermal heating systems have the potential to displace the use of gas-fired space

and water heating requirements. On pages 13 and 14 of his direct testimony, intervenor Wyman alleges that the installation of such district geothermal systems by utilities "fits within existing and anticipated law and policy."

Q. Are there currently legal or statutory issues which preclude gas utilities from transitioning to thermal companies as proposed by witnesses Rodriguez and Wyman?

A. Yes. Staff Counsel has advised us that there are legal issues which prohibit gas corporations from transitioning to thermal companies. Staff Counsel has advised us that they will address such legal and statutory issues in briefs as part of these proceedings.

Q. What does the Panel recommend with respect to witness Rodriguez's and intervenor Wyman's proposals?

A. We recommend that the Commission reject such proposals.

District Heating Study

Q. Provide a summary of the City Policy Panel's proposals regarding its proposed District

Heating Study.

A. On pages 26 through 28 of its direct testimony, the City Policy Panel proposes that the Companies conduct a four-phase feasibility study to examine, pilot, and report on the potential for utility-owned district geothermal systems to replace the need for leak-prone pipe (LPP) replacement capital projects. Phase one of the study would require that the Companies hire a consultant to examine the feasibility of deploying district energy systems, identifying and targeting areas of high LPP concentrations. Phase two of the study would consist of selecting ten to twenty high-potential or unique use-case sites for further detailed assessment and design, ultimately ranking such sites based on potential benefits, costs, and feasibility. Phase three would consist of a pilot program to build and test district energy systems at several of the sites identified and examined in phases one and two. Phase four of the study would assess the scalability of the pilot locations, examine the costs and benefits of expanding the pilot, and would identify required

changes to the pilot structure and regulatory construct needed to increase adoption of district energy systems.

Q. Did the City Policy Panel provide a forecast of District Heating Study costs, or propose a way to pay for the study?

A. While the City Policy Panel did not provide a forecast of the costs related to the study, it did propose that the Commission consider changes to the regulatory model to allow the avoided costs of LPP replacements to be used to offset the capital costs of district energy systems.

Q. Does the Panel agree with the City Policy Panel’s proposed District Heat Study?

A. Partially. While we agree that displacing LPP replacement capital projects is attractive as a potential area where gas capital expenditures could be avoided using geothermal heat pumps or other technologies, we disagree with the City’s implication that utility ownership of shared loop fields should be the default solution.

Q. Why should utility ownership of shared loop fields not be the default option?

A. First, as noted earlier, Staff Counsel has

advised us that there may be legal issues with utility ownership of non-gas assets. Second, setting aside those legal issues for the sake of argument, we see utility ownership of geothermal loop fields and/or heat pumps as a last resort in the event that the private market is unwilling or unable to provide a cost-effective solution. Whether market conditions warrant utility ownership of geothermal systems should be tested under the same criteria as utility ownership of any other type of distributed energy resource (DER).

Q. Has the Commission adopted a set of criteria for utility ownership of DER?

A. Yes, the Commission's February 26, 2015 Order Adopting Regulatory Policy Framework and Implementation Plan in Case 14-M-0101 (Track One Order) provides such criteria. On page 68 of the Track One Order, the Commission establishes that utilities will not be allowed to own DER assets as a general rule, unless markets have had an opportunity to provide service and have failed to do so in a cost-effective manner. On page 70 of the Track One Order, the Commission

lays out four exceptions to the general rule under which utility ownership would be allowed where: (1) procurement of DER has been solicited to meet a system need, and the utility has demonstrated that competitive alternatives proposed by non-utility parties are clearly inadequate or more costly than a traditional utility infrastructure alternative; (2) a project consists of electric energy storage integrated into electric distribution grid infrastructure; (3) a project will enable low and moderate income (LMI) customers to benefit from DER where markets are not likely to satisfy the need; and (4) a project is part of a Demonstration Project.

Q. Has the City Policy Panel demonstrated a market failure, or that its proposed District Energy Study and pilot are exempt from the general rule prohibiting utility ownership of DER?

A. No, it has not.

Q. Should the District Energy Study be considered a Demonstration Project?

A. No. The Track One Order provided Demonstration Project authorization and budgets for Electric

utilities only. As a procedural matter, utility Demonstration Projects require Department of Public Service Staff approval to move forward – clearly, by our testimony on this matter, approval by Staff of a utility-owned district energy system is by no means guaranteed or even likely. Further, the Companies are already operating one utility-owned shared geothermal loop field as a Demonstration Project approved by the Commission in the Companies' last rate proceeding – a second such demonstration project is not needed. Furthermore, our rationale for not supporting the Companies' proposals to expand that Demonstration Project as expressed in our direct testimony apply equally to the City's proposal.

Q. Does this conclude your testimony at this time?

A. Yes.